A Clinician-Guided Framework for Endoscopic AI: Developing PanEndoAtlas and Benchmarking Foundation Models Across the Full GI Spectrum

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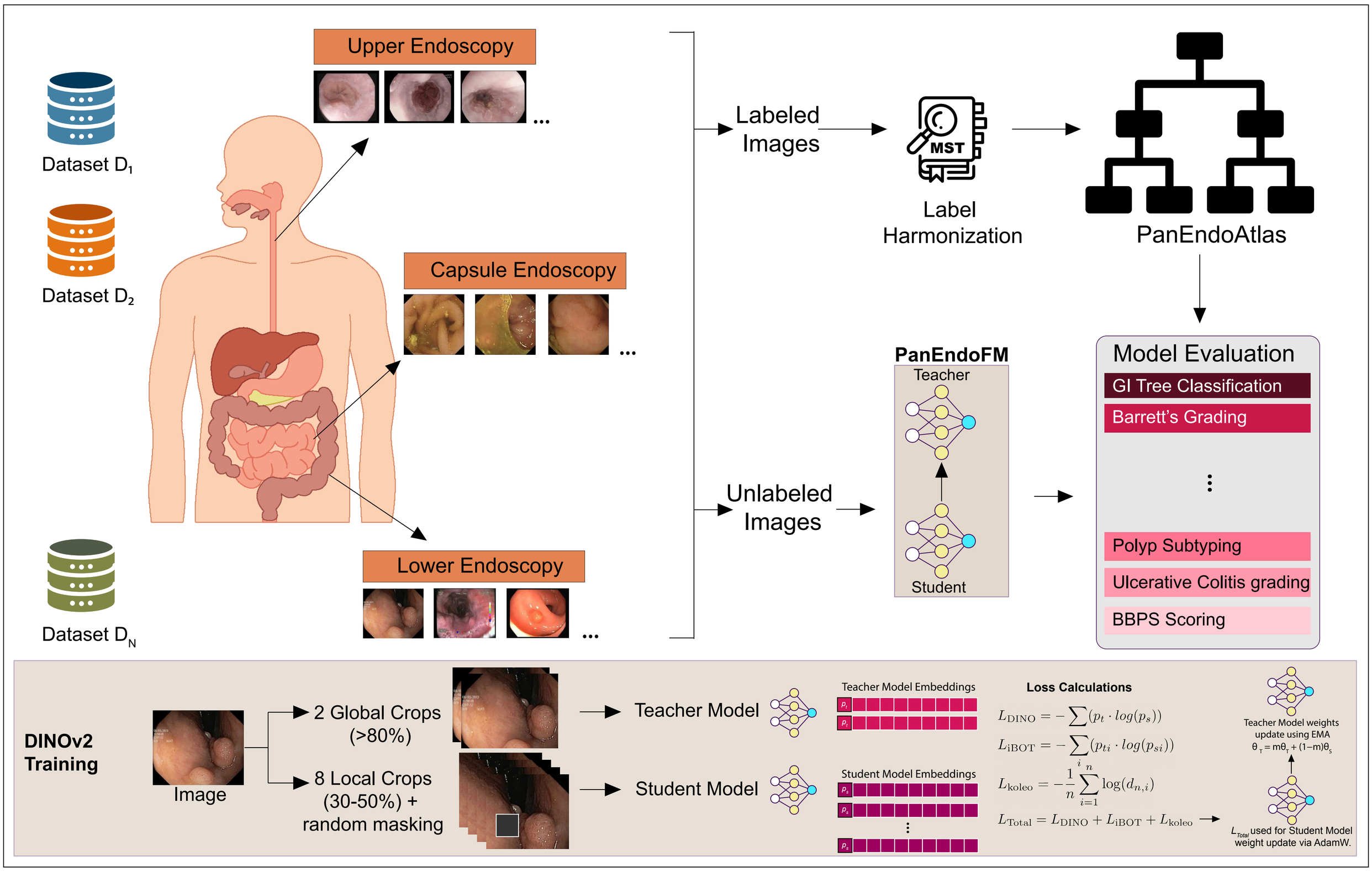
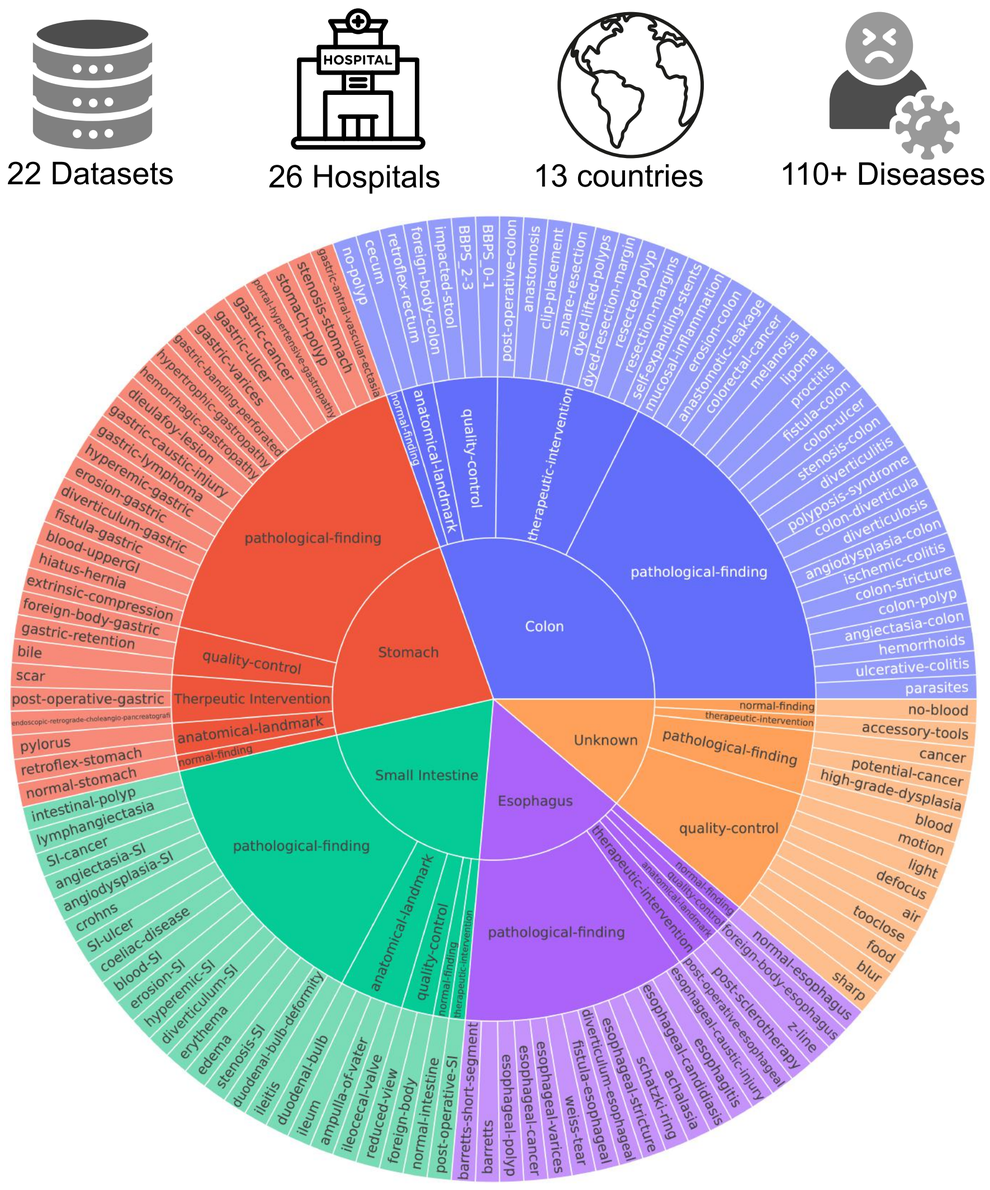


PanEndoAtlas



>420K Labeled Images

Hierarchical structure enables models to leverage shared features across related conditions, improving performance on long-tail of rare diseases.

Results

Task	Classes	PanEndoFM	EndoFM-LV	EndoSSL	ViT-B/16	ResNet-50
GITree	111	**0.648**	***0.570***	0.551	0.500	0.515
Barrett's Grading	4	**0.775**	***0.733***	0.581	0.681	0.707
UCC Grading	3	**0.571**	***0.565***	0.507	0.550	0.525
Polyp Subtyping	4	0.575	0.500	**0.704**	0.542	***0.643***
BBPS Scoring	3	0.809	**0.913**	0.524	***0.830***	0.685
Esophagus Diseases	13	**0.772**	0.590	0.550	***0.682***	0.446
Stomach Diseases	14	**0.539**	***0.539***	0.521	0.438	0.498
SI Diseases	20	***0.594***	0.481	0.591	**0.597**	0.528
Colon Diseases	30	***0.552***	**0.615**	0.455	0.494	0.503
Stomach Landmark	3	**0.950**	0.869	0.788	***0.879***	0.820

PanEndoX Benchmark: Macro-AUCs reported on test set. Highest value is in **red**, second highest is in ***italics***.

- PanEndoFM outperforms models in **comprehensive disease classification** across GI track (GITree classification).
- PanEndoFM, trained on 10 million unlabeled images from publicly available datasets, outperforms existing endoscopy foundation models on **6/10 tasks** focused on disease identification.
- Targeted enrichment of specific GI conditions in the pretraining dataset (polyps in EndoSSL) leads to superior performance on the GI condition but poor generalization to other diseases.

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